15.4.3 Pavement Thickness

A thorough soils investigation should be conducted to determine the strength of the existing soils and drainage characteristics. The investigation and design recommendations should make sure that the project is not being built on unstable soils (such as expansive clays or collapsible soils) that will cause significant problems. These must be dealt with during design and construction. The drainage of the site must be thoroughly evaluated because saturated soils can have a dramatic effect on the load-carrying capacity of the pavement. When areas of poor soils are encountered, their removal and replacement with a higher-quality material should be considered.

For bicycle/walking paths, the type of traffic that will operate on the surface should be considered. Periodic vehicular traffic, such as emergency, patrol, snow removal, maintenance, or other vehicles, must be considered in the design. For most paths, full-depth asphalt provides an excellent surface because of its speed of construction. The thickness of the HMA surface will vary from 3 to 5 inches (75 to 125 millimeters) depending on the traffic and the soil conditions of the site. It may be desirable to place a base course of crushed stone or reclaimed asphalt pavement to help distribute the load or as a construction aid to ease the grading and layout of the path.

Tennis courts should generally have 3 to 4 inches (75 to 100 millimeters) of HMA placed on a 4- to 8-inch (100 to 200 millimeters) layer of compacted base course (consisting of crushed aggregate or reclaimed asphalt pavement). It is also recommended that the subgrade be leveled and compacted to provide a finish elevation of 4 to 6 inches (100 to 150 millimeters) above the surrounding ground surface. This will provide for water runoff and vegetation control around the perimeter.

Asphalt Mix Design

The type of mix to be used for bicycle/walking paths and other recreational facilities should be different than that used for parking lots or highway facilities. The mix-design criteria used for asphalt pavements designed for vehicular traffic will not, in most cases, provide a mix of the needed durability and workability required for recreational facilities. A mix developed for recreational use will generally contain more asphalt binder, have less air voids, and be less prone to oxidation, raveling, and cracking than a typical roadway mix. The objective of the asphalt mix is to provide a cost-effective mix that will have the following characteristics:

- Sufficient asphalt to provide for a durable surface
- A smooth uniform surface texture
- Sufficient workability to allow ease of handling during construction and proper compaction

Hot mix asphalt for recreational facilities should have a fine-graded aggregate blend with a nominal maximum size of ½ inch (12.5 millimeters) or less. For some applications such as tennis courts, it is recommended that a nominal maximum size of ⅜ inch (9.5 millimeters) or less be considered. Aggregates used in surface mixes should be free from impurities such as

iron pyrite that can cause staining on the surface of the finished project. The mix can be designed using the Superpave design procedures discussed in Chapter 4, "Hot Mix Asphalt (HMA) Design." But, rather than having a design air-voids content of 4 percent, these mixes should be designed having a design air-voids content of 2–4 percent. This will allow more asphalt in the mix, thereby improving durability.

The asphalt binder grade chosen for a recreational facility should be based on the climate conditions. A standard, unmodified asphalt binder can be used. It is suggested that the cold temperature properties of the selected binder be one grade lower than that used for highways (if available locally). For example, if a PG 64–22 binder is used for highways in a region, a PG 64–28 or PG 58–28 would be appropriate for the recreational facility.

Construction Considerations

Before construction, all vegetation and all unsuitable soils should be removed and the subgrade should be scarified and compacted to at least 95 percent of modified proctor density (ASTM D 1557/AASHTO T 180). After compaction, the subgrade should be proof-rolled with a heavy pneumatic roller or dump truck to ensure that there are no soft spots. Any soft spots identified should be removed and replaced with a stable material. After compaction, the subgrade should be treated with an herbicide. This must be done carefully to insure that surrounding vegetation is not adversely affected. If an aggregate base course is used, it should also be compacted to at least 95 percent of modified proctor density.

For bicycle/walking paths, the asphalt layer can be placed in one lift. To enhance surface smoothness on tennis courts and running tracks, an asphalt layer placed in two equal lifts is recommended. Most tennis court specifications require that tolerance not vary more than ⅛ inch in 10 feet (3 millimeters in 3 meters) when measured in any direction.

15.5 Asphalt Roofs

15.5.1 History

Roofing was probably the world's first craft because shelter has always been a primary need of people. Asphalt, with its extraordinary combination of waterproofing, preservative, and cementing qualities, was an early discovery of ancient people and has been used extensively in the construction of their buildings and roads.

Today, asphalt is recognized more than ever as an outstanding roofing material and is used extensively to cover all kinds of buildings, from homes to factories. Asphalt roofing materials and construction methods are fully covered by specifications of the American Society for Testing and Materials (ASTM) and many government and private agencies. In addition, all types of asphalt roofing are sanctioned by federal and other public agencies.

Asphalt roofs may be built in single- or multiple-layer systems with impregnated fabrics or with manufactured shingles.

15.5.2 Types of Asphalt Roofing

Prepared Roofing

Asphalt shingles. This type is composed of three basic materials—asphalt, felt, and mineral granules. The felt is impregnated with asphalt. Both sides of the saturated felt are then covered with a harder, tougher coat of asphalt, in which are embedded mineral granules. This composition is then cut into individual shingles or strips. Asphalt shingles are available in almost any color from pastels to deep rich tones. Factory techniques for the application of granules have improved with the development of new permanent colors, and granules may be applied in almost any blend or pattern of colors desired.

Asphalt roll roofing—smooth-surfaced. This type consists of a single layer of roofing felt saturated with asphalt and coated on both sides with a harder asphalt.

Asphalt roll roofing—mineral-surfaced. This type consists of smooth-surfaced roll roofing in which mineral granules have been embedded on either one or both sides.

Prepared roofings are those products manufactured and packaged ready to apply to the roof deck, usually by nailing only. Asphalt shingles and certain roll roofings are in this category. They consist of a base of saturated felt to which has been added an asphalt coating and, usually, a surfacing of mineral granules.

Of the types listed, the shingle roof has been the most common. Asphalt shingles are applied to the steep roofs of homes, barns, and similar structures. They fill the need for attractive, efficient, but low-cost roofing for residential construction.

Asphalt Built-Up Roofing

An asphalt built-up roof is composed of several layers of asphalt-saturated felt applied principally to a flat-roof deck. Asphalt is applied between the felt layers for adhesion. Over the top layer, a flood coat of asphalt is added and then covered with gravel or slag.

The built-up roof is a contrasting type, constructed on the flat or nearly flat tops of very large buildings—industrial, public, and many others.

For built-up roofing, only the components are manufactured and delivered to the job. The roofing itself is built up directly on the roof deck in a series of layers or plies, sandwiched between layers of asphalt. A cover of aggregate is then applied over the final layer. The mopping on of the asphalt and the spreading of slag or gravel may be accomplished with the most readily available and simplest equipment. All operations can be performed quickly by hand application, although mechanical felt-laying machines, hot-asphalt applicators, and gravel spreaders will do the job more easily and rapidly on very large roofs.

The heart of the built-up roofing operation is the kettle for heating the asphalt to the proper application temperature. For efficient operation, the kettle must be able to bring the asphalt up to this temperature in a short time without overheating at the start—or at any time during the kettle's operation. Overheating may seriously damage the asphalt and considerably reduce the life expectancy of the roof. Overheating may also cause the asphalt to be applied too thinly, resulting in poor adhesion between the roofing felts and possible later separation of the plies.

Properly constructed, an asphalt built-up roof will remain in service condition for many years with little or no maintenance required.

Either prepared or built-up roofing can be used on many roofs, but the choice is limited somewhat by the slope of the deck to which it is to be applied. Built-up roofs are usually applied only on the flatter slopes, while the application of asphalt shingles and other prepared roofing is typically used for the steeper slopes.

15.6 Pavement Noise Mitigation

15.6.1 Introduction

Research in Europe and in the United States has shown that it is possible to build asphalt pavement surfaces that will provide reduced noise levels. Throughout the world, sound caused by transportation systems is the number one noise complaint. Noise generated by traffic is one of the prime offenders. Engine (power train), exhaust, aerodynamic, and tire/pavement noise all contribute to traffic noise. Modification of pavement surface type and/or texture can result in significant tire/pavement noise reductions. A quiet road surface can be built while at the same time considering safety, durability, and cost using one or more of the following approaches:

- A smooth surface texture using small top-size aggregate
- An open-graded friction course (OGFC) with a high air-voids content

15.6.2 Nature of Noise

Noise, like all other sounds, is a form of acoustic energy. It differs from pleasant sounds only in that it often disturbs us and has the characteristics of an uninvited guest. Understanding noise requires a basic knowledge of sound physics and how humans respond to it.

Sound is fluctuation of air in a wavelike motion. Associated with this motion is acoustic or sound pressure that is measured in decibels. The decibel combines the magnitude of sound with how humans hear. Because human hearing covers such a wide range of sounds, it does not lend itself to measurement with a linear scale. If a linear scale were used, most sounds occurring in daily life on a scale of 0 to 1 would be recorded between 0.00 and 0.01. Thus, it would be difficult to discriminate between sound levels in our daily lives.

Instead of a linear scale, a logarithmic scale is used to represent sound levels and the unit is called a decibel or dB. The term dB(A) is used when referring to the sound levels that have been A-weighted. The ear has its own filtering mechanisms, and the inclusion of the A after dB indicates that the scale has been normalized or "fine tuned" to hear like a human. The decibel scale ranges from 0 dB(A), the threshold of human hearing, to 140 dB(A) where serious hearing damage can occur. Table 15.9 represents this scale and some of the levels associated with various daily activities.

Alongside a freeway, the sound level (i.e., noise) might be in the range of 70–80 dB(A). As a rule, an individual can differentiate between two levels of similar sound that are at least 3 dB(A) different in level.

Table 15.9
Noise Levels Associated with Common Activities

Activity	Noise Level dB(A)
Lawnmower	95
Loud Shout	90
Motorcycle Passing 50 Feet Away	85
Blender At 3 Feet	85
Car Traveling 60 Mph Passing 50 Feet Away	80
Normal Conversation	60
Quiet Living Room	40
Peaceful Neighborhood Setting	40–50
Serene Farm Setting	30

Noise from traffic is due to a large number of moving noise sources spaced along an entire roadway. This type of noise is classified as a line source because noise is transmitted along the entire roadway. As the distance from the line of source increases, the noise level decreases at a lower rate than it would from a single-point noise source. This is because the noise from the more distant sources in the line begins to contribute more noise relative to that of the closest individual point source. The net result is that for line sources, a doubling of the distance results in a 3 dB(A) reduction in the noise level instead of 6 dB(A). Thus, if a point that is 5 meters from the center of the noise source (the center of the lane) has a noise level of 85 dB(A), then a point 10 meters from the edge of the roadway would have a noise level of 82 dB(A). This is illustrated in Figure 15.14.

15.6.3 Noise Characteristics of HMA Pavements

The proper selection of the pavement surface can result in a significant reduction in the noise level alongside the roadway. The selection of pavement type and the maximum gradation of the aggregate will affect the noise level. Selection criteria are discussed in the following paragraphs. As a pavement ages, its noise level increases because of changes in the surface texture. The HMA surface may start to lose particles, crack at the surface, and in the case of an OGFC, result in clogged pores from accumulated dust or deicing materials.

Dense-Graded Mixes

The National Center for Asphalt Technology (NCAT) has tested 90 in-service dense-graded HMA mixes of all ages, throughout the United States using a close-proximity (CPX) noise trailer. For these mixes, the noise level has ranged from 93.9 dB(A) to 101.8 dB(A). (The noise level measured

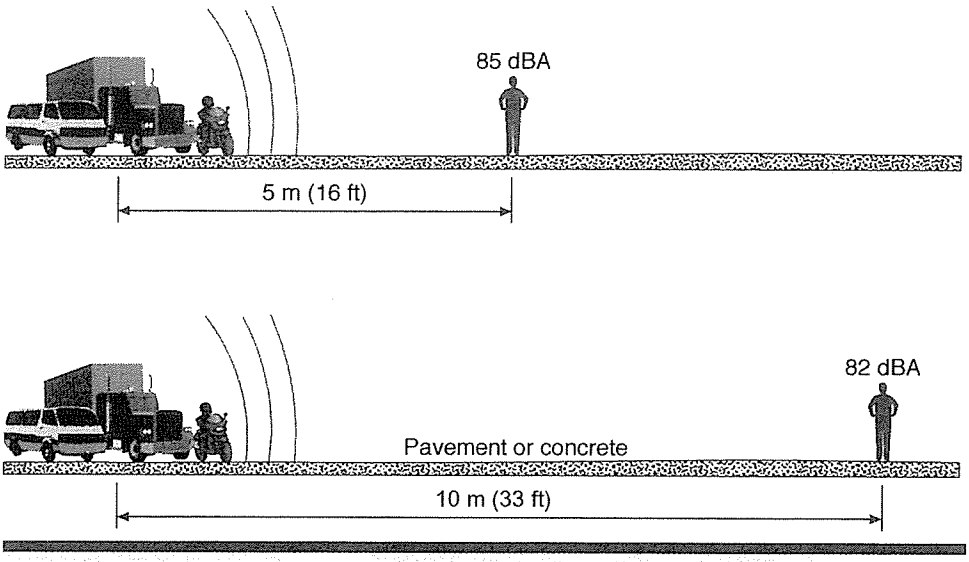


Figure 15.14 Effect of Distance on Traffic Noise over a Paved Surface

25 feet from the roadway edge will be about 23 dB(A) less than this value taken from the CPX trailer.) It was found that as the surface texture of the dense-graded mix increases, the noise level increases. The following shows average noise levels for different maximum aggregate size mixes from the NCAT test program:

- ¾ inch (19 millimeters)—96 dB(A)
- ½ inch (12.5 millimeters)—95 dB(A)
- ⅜ inch (9.5 millimeters)—93 dB(A)

Stone Matrix Asphalt Mixes

Stone Matrix Asphalt (SMA) mixes are seeing increased use in the United States to reduce rutting or permanent deformation of HMA mixes. They were originally developed in Europe. Research in Europe has shown that these surfaces can have a noise level of 96.3–100.6 dB(A) when measured with a CPX trailer. The noise level increases as the maximum-size aggregate increases. The following shows the average noise levels for different maximum-size SMA mixes from the NCAT test program:

- ¾ inch (19 millimeters)—97 dB(A)
- ½ inch (12.5 millimeters)—95 dB(A)
- ⅜ inch (9.5 millimeters)—93 dB(A)

Open-Graded Friction Courses

Open-Graded Friction Course (OGFC) surfaces typically have lower noise levels than dense-graded HMA surfaces. An OGFC can reduce the noise level 3–5 dB(A), when compared with nonporous HMA. This is accomplished because the air voids in a pavement provide a means for air trapped between the tire and the pavement surface to escape, which provides for increased sound absorption. To achieve this improved noise performance, the voids need to be interconnected. The added advantage

of these surfaces is that they also reduce splash and spray and increase the skid resistance of HMA wearing courses. The noise level from an OGFC surface depends on four factors: the interconnected air voids or permeability of the surface, the thickness of the OGFC layer, the gradation (the maximum size of the aggregate) of the OGFC, and the quantity of the binder used in the OGFC mixture.

The optimum surface for noise reduction in an OFGC is a mix with more than 15 percent air voids. The increase in air voids results in improved slash-spray and hydroplaning reduction characteristics of the pavement surface. The mix gradation will affect the noise level of the OGFC. As the gradation gets coarser, the noise level increases at low frequencies. Research has shown that the use of a coarser aggregate for an OGFC will increase the megatexture of the pavement and thus increase the noise level at the lower frequencies. The asphalt binder generally will have little effect on the noise level of an OGFC pavement. But studies conducted by the Arizona Department of Transportation have shown that the use of a crumb-rubber modified-asphalt binder can reduce the noise level an additional 1–2 dB(A).

15.7 Miscellaneous Asphalt Uses

Some types of asphalt construction require special consideration of either the construction methods or the composition of the asphalt mixture. Asphalt as an engineering material is versatile and, when designed properly, is ideal for many forms of paving and other applications.

15.7.1 Bridge-Deck Paving

General

Asphalt surfaces are desirable for bridge decks because they are not affected by salts used for de-icing, and if properly constructed they may waterproof the bridge floor, thus preventing water and salts from damaging it. Asphalt surfaces can also be highly skid-resistant, providing a safe bridge deck.

Asphalt Mixes

Dense-graded asphalt mixtures are recommended for leveling and intermediate courses. In areas where chemicals are expected to be used to de-ice the pavement, aggregates that do not react with de-icing chemicals should be selected. A lightweight aggregate may be desirable to reduce the weight of the asphalt pavement on the bridge.

Both dense-graded and stone matrix asphalt HMA mixtures have been used for bridge decks. In many cases on special projects, very stiff polymer-modified asphalt binders (up to PG 82–22) have been used to prevent rutting or other forms of permanent deformation in that HMA layer. Also specially designed asphalt waterproofing membranes have been placed on the deck before the HMA layer is placed.

15.7.2 Construction Methods

The surface to be paved, whether it is a PCC or a corrugated steel bridge deck, must be cleaned of all loose and foreign matter. Power brooms and compressed air tools may be necessary to accomplish this task. The surface of the PCC deck is then given a tack coat of SS-lh or CSS-1h asphalt emulsion diluted with an equal amount of water and applied at a rate of 0.05–0.20 gallons per square yard (0.2–0.9 liter per square meter).

After mill scale is removed, steel decks usually require painting with a heavy asphalt paint. Vertical surfaces are also painted to a depth greater than the thickness of the subsequent paving. The tack coat is scrubbed in, thus providing a continuous asphalt film with no holes in the coating.

A leveling course of dense-graded HMA is laid and compacted. For steel decks, this course should have a compacted depth extending about ½ inch (13 millimeters) above the corrugations. Rolling the leveling course on a steel deck is done in a direction parallel to the corrugations.

The next step is to apply a tack coat and then place and compact the wearing course.

Pneumatic rollers having 80 psi (550 kilopascals) tire contact pressure are used for rolling all courses but finish rolling is typically done with a steel-wheel roller. Inaccessible areas require hand or small vibratory compactors.

15.7.3 Industrial Floors

General

Asphalt floors are particularly adaptable for use in many factories and warehouses. Above ground level, they are ordinarily laid on PCC decks. At or below ground level, they may be designed and constructed similarly to roadways or parking lots. Asphalt floors are waterproof, highly resistant to wear, and dustless. The use of a highly siliceous mineral aggregate in the mix may make it resistant to certain dilutions of acid.

Asphalt Mastic Floors

Originally, an asphalt mastic was prepared using crushed-rock asphalt. The term now generally applies to a mixture of graded sand, mineral filler, and asphalt cement quite similar in character to sheet-asphalt paving. Factory and warehouse floors are frequently subjected to very heavy unit loads, carried on small-diameter narrow metal wheels. When confronted by this condition, the regular sheet-asphalt mixture is modified by using a harder asphalt binder (such as a PG 70–22 or PG 76–22) and a higher proportion of very fine mineral filler. A maximum thickness of less than 1 inch (25 millimeters) is typically recommended.

Asphalt Floors

Dense-graded asphalt such as that used for paving highways and airports is particularly well adapted for paving warehouse and factory floors. Asphalt itself is unaffected by acids, but the mineral aggregate should be acid-resistant. Care should taken to design and compact the mixture to produce high density and waterproofing.

15.7.4 Roof Deck Parking

General

The roofs of many modern structures serve as parking areas. Asphalt paving is particularly adaptable to roof parking lots for reasons similar to those for bridge-deck paving.

The importance of proper drainage cannot be overemphasized. To ensure rapid runoff of water to drains, the surface should be pitched or graded at a 2 percent slope or ¼ inch per foot (20 millimeters per meter).

Asphalt pavement transfers large moving and static loads to the roofing membrane. Any lateral movement in the membrane causes the pavement to move. Any differential heaving under an asphalt pavement under load causes distress or failure to the underlying membrane. Also, any excess or free asphalt in the membrane is absorbed into the wearing surface and causes it to soften and possibly bleed.

If an asphalt pavement is placed over a deck area constructed of prestressed concrete beams, reflection cracks are likely to come through the pavement at the junction between successive beams. To minimize reflection cracking, some type of stress-absorbing membrane can be placed over each joint. Such a membrane also helps provide a tighter seal against the passage of water.

The two types of roof-deck paving are (a) slab and pavement with a built-in slip plane and (b) slab and pavement bonded together.

Free-Movement Plane Construction

Free-movement plane construction provides a distinct, free-movement plane between the structural roof slab and the pavement.

A positive bond breaker, such as a lime or fine sand, is spread over the concrete deck to allow freedom of movement between the concrete deck and the pavement. A conventional built-up roof consisting of four or five plies of lightweight roofing felt is placed on the roof. However, the final flood coat and aggregate cover is not applied. In its place, a very light tack coat of diluted SS-1h or CSS-1h asphalt emulsion is applied to the top felt layer. On top of this, an HMA layer is placed and compacted. A minimum thickness of 2 inches (50 millimeters) is recommended, and a very densely graded mix having a maximum aggregate size of about ⅜ inch (9.5 millimeters) should be used.

Bonding Roof Deck and Pavement

To bond pavement to a roof deck after cracks and joints have been sealed, a conventional four- to five-ply built-up roof is applied. Each layer, except the surface of the top layer, is spot mopped with asphalt. No mopping is done within 15 inches (375 millimeters) of a PCC slab joint. The top layer is then covered with 2–3 inches (50–75 millimeters) of a dense-graded HMA. Another method is to use an asphalt membrane that contains mineral filler and fibers for body and strength. The membrane is carefully spread to ensure a uniform thickness. Often, a protective board ⅛–¼ inch (3–6 millimeters) thick is placed over the membrane. The membrane or board is overlaid with one or two lifts of HMA having an asphalt content higher than normal. The total thickness of the mix should not exceed 3 inches (75 millimeters). The aim is to obtain an impervious mixture that weathers well and has high stability or strength.

Small rollers or sidewalk rollers are often used because of the restricted working area and the maximum allowable loads for the building. Small hand-operated electric or gasoline-powered vibratory compactors are used to work around areas inaccessible to rollers.

15.8 Leaching Pads

Because asphalt is an inert material, it can be used to construct all sorts of structures for handling very corrosive materials.

15.8.1 Uses of Leaching Pads

Leaching pads are used to extract metals from low-grade ores, such as copper, gold, etc. Acid solutions are sprayed over stockpiles of the ore, and the solutions leach out the metals. The metals are then recovered from the leachates.

15.8.2 Design of Leach Pads

Leach pads are designed using hot mix asphalt with a high binder content of a soft grade to assure a flexible pavement capable of withstanding some subgrade settlement.

To ensure that the pad pavement is impermeable, leach pads are sealed with a high softening point (polymer-modified) asphalt.

Also included in the design are dikes, channels, and retention ponds for gathering and holding the leachate.

15.8.3 Materials

Mix Binder—The binder selection should be one or two grades softer than what would usually be used for asphalt pavements in the climatic area where the pads are to be constructed. Example: If a PG 70–10 is the normal paving grade used, then select either a PG 64–10 or a PG 58–10.

Mix aggregate—Use a ¾-inch (19 millimeters) nominal-size aggregate, with a dust-to-binder ratio of at least 1, and design for 0 air voids.

Membrane seal—Use a 170/200 pen or high softening point material that is applied at the rate of 0.35 gallon per square yard (1.10 liters per square meter).

15.8.4 Construction of Leaching Pads

The following steps should be followed in constructing a leaching pad.

1. Prepare the subgrade and compact to a uniform density.
2. Place 4 inches (100 millimeters) of the hot mix asphalt and compact to 96 percent of maximum specific gravity.

3. Apply the membrane seal in three applications to eliminate the pin holing effect of this type of material.
4. With front-end loaders or bulldozers, push a 3-foot (1 meter) layer of the material over the pad as a cushion to avoid the possibility of puncturing the pad.

15.9 Environmental Applications of Asphalt

15.9.1 General

For decades, many civil engineering solutions to environmentally sensitive projects have incorporated asphalt. It has been used because it is flexible, strong, chemically stable and nonreactive, impermeable, recyclable, economical, and readily available. All sorts of asphalt materials are used in environmental applications: sprayed-on membranes of emulsified asphalt, asphalt cement, and modified asphalt, dense-graded HMA, open-graded HMA, hydraulic mixes, sand mixes, and mastic asphalt liners.

Contrary to the fears of some environmental groups, scientists have established that asphalt, in the form of an emulsion or incorporated into a mixture of aggregates, does not leach into water, contaminate soil, or kill fish. The results of leachability tests of new HMA show very low levels of leachable compounds, well below any current Environmental Protection Agency guidelines. Scientists have proved that asphalt is an environmentally safe material.

15.9.2 Examples of Environmental Uses of Asphalt

1. **Fish hatcheries.** Oregon and Washington have more than 35 asphalt-lined fish hatchery ponds. These ponds are used to raise fingerling fish until they are large enough to be released. A typical hatchery pond is 5–7 feet (1.5–2.1 meters) deep and covers about a half-acre (0.2 hectares). The pond liner is constructed of 2 or 3 inches (50–75 millimeters) of HMA over 8 inches (200 millimeters) of aggregate base course. Most liners are sealed with an asphalt emulsion so that bacteria cannot inhabit the tiny surface irregularities in the HMA liner.
2. **Water reservoir liners.** Throughout the world, many asphalt-lined water reservoirs have been constructed over the past 100 years. The design of a reservoir liner must address permeability, durability, flexibility, resistance to weathering, the ability to absorb stresses during filling and drawdown cycles, long-term loading stresses caused by settlement in the supporting subgrade, and the weight of maintenance vehicles when the reservoir is periodically cleaned. The composite liners required for large drinking water reservoirs are typically designed using both dense-graded and open-graded HMA. Dense-graded mixes provide an impermeable

barrier. Open-graded interlayers are used in the liner structure to monitor and intercept any leakage that may occur. Figure 15.15 shows a typical asphalt-lined reservoir.

3. **Landfill caps.** HMA can be used to protect the environment by providing impermeable caps for abandoned landfills and pads for storing hazardous or waste materials. These caps or pads typically consist of hydraulic (impermeable) asphalt, paving fabric, and aggregate base. Many caps are designed to support haul traffic. Tacoma, Washington, covered three acres of its landfill for use as a solid-waste processing station. The cap consisted of 12 inches (300 millimeters) of sand to cover the refuse, a high-density polyethylene membrane, and then 12 more inches (300 millimeters) of sand to protect the membrane from construction equipment. An 8-inch (200 millimeters) layer of dense-graded aggregate was placed over the sand and surfaced with 3 inches (75 millimeters) of HMA.

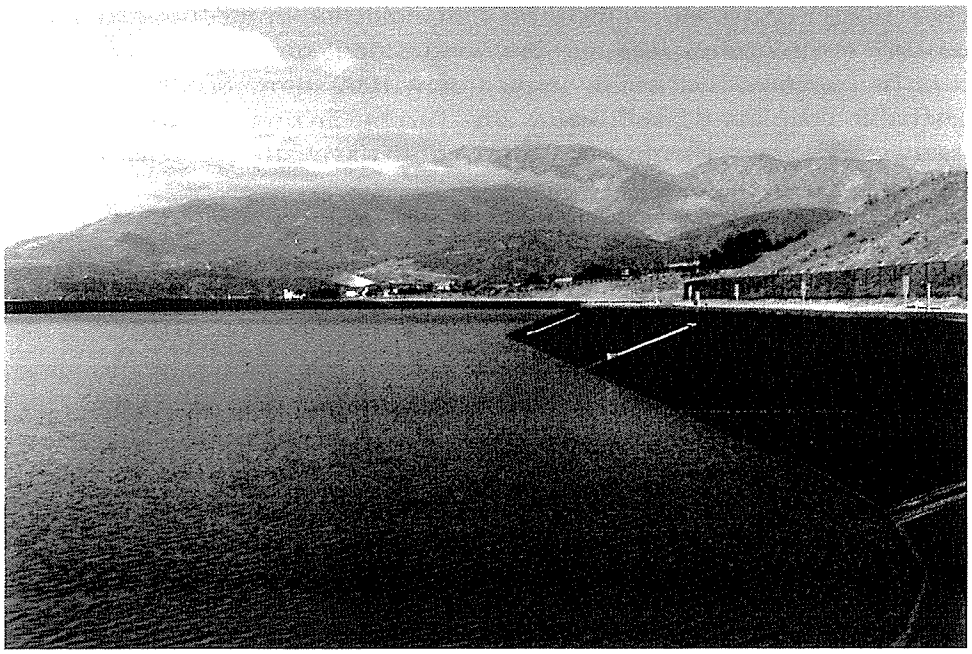


Figure 15.15 A Typical Asphalt-Lined Reservoir

15.10 Uses of HMA for Race and Test Tracks

15.10.1 Introduction

Whether a racetrack or a test track, a track almost certainly will be paved with HMA—the only material that can consistently provide the smooth surfaces necessary for vehicles traveling at such high speeds (see Figure 15.16). HMA pavements can be constructed to smoothness standards unmatched by any other surfacing type.

Each track has its own unique characteristics as to length, shape, and banking angles. Racetracks are usually oval in shape but can be quad- or tri-ovals and have banking angles that range from 12 to 36 degrees. Test tracks, on the other hand, usually are oval; however, some older tracks are circular in shape. Slopes on test tracks can range from the normal 2-percent cross-slope typical of a normal pavement, to as much as a 70-degree cross-slope. The greater slopes are needed to provide neutral steer though the turns while vehicles are tested at high speeds.

Test facilities not only have high speed tracks but generally incorporate road courses, stability testing pads, and/or segments of pavements that duplicate existing pavements. These test facilities are a way to compare how vehicles react to different pavement conditions.



Figure 15.16 HMA-Surfaced Race Track

Pavements for test tracks and racetracks are constructed in all sorts of climate conditions, and these conditions must be considered during pavement design and construction. These tracks provide very wide, paved surfaces, which make them susceptible to thermal cracking. The pavement is subjected not to heavy loads but to the scrubbing action of the side thrust of the tire as the vehicles go through the turns of the track. Therefore, track pavements need to be stable and durable.

Drainage is another important factor. The pavement design should include a drainage layer within the pavement structure. Special attention must be given to the areas behind the crash wall and under the stands. Any water from these areas must be intercepted.

15.10.2 Materials Requirements

Binders

The selected binder should be at least two PG grades higher than that commonly used for the area where the track will be constructed. The binder should have a minimum softening point of 180°F (82°C).

The selected binder should be two PG grades higher on the high-temperature side and can be one grade less to minimize thermal cracking; the low-temperature grade should be one or two grades lower than that usually selected, compared with a normal highway binder grade in that climate. For instance, instead of using a PG 64–22, a PG 76–28 or PG 82–28 should be considered.

Binders for the underlying layer of the pavement structure can be the same unmodified binder as would be selected by the local DOT for that region of the state where the track is located.

Aggregates

The aggregates used in track surfacing must be a hard, durable material that provides good surface friction. Granite and slag make good aggregates; however, other sources of hard, non-polishing, durable aggregates can be used. Very hard, angular aggregates should be avoided because they tend to shred rubber from tires, resulting in slower races with more frequent pitstops for tire changes.

The same aggregates used for highway construction can be used for the underlying layers of the pavement structure.

Mix Design

Mix designers should consider high stability and high density as their main objectives. Stone matrix asphalt mixes with their high binder content and thicker mortar should be considered when the local paving industry has experience using this type of mix regularly. The Indianapolis Speedway used an SMA mix.

A 75-blow Marshall or a minimum 100 gyration Superpave mix design is appropriate for race track mixes. A gradation of ¾ inch (19 millimeters)

could be used for underlying layers of the pavement structure; however, for the surface course a gradation of ⅜ inch (9.5 millimeters) should be used.

Field compaction requirements should be 96 percent of maximum specific gravity of the design mix.

15.10.3 Construction Practices

Race and test track construction is very complex construction work. This is especially true for the paving contractor. The contractor not only deals with closer tolerances, a smoother surface, and higher densities, but also must accomplish those tasks while working on steep slopes. Construction work certainly tests the ingenuity of the contractor's personnel.

Anyone contemplating being involved in race or test track design or construction should consult with a person who has previously built a track.

For slopes 18 degrees or less, paving procedures are similar to normal highway paving. On slopes between 18 and 24 degrees, the rollers need lateral support, which can usually be accomplished from the lower side (Figure 15.17). Care must be taken to ensure that the lower portion of the roller does not dig into the mix. Material transfer vehicles are needed to deliver the mix to the paving machine. Adjustments must be made to the paving machine, to ensure proper delivery of the mix to the screed. Once the slope is steeper than 20 degrees, all the equipment from the top or bottom of the track must be anchored. Crawler tractors, backhoes, or cranes make good deadmen that can move along the top of the track at the paving speed. Heavily weighted crawler-mounted machines connected by I-beams of different lengths can be used to anchor the paver from the bottom of the slope. Large crab-steering, shoulder-widening machines can be used in the same configuration to hold up compaction rollers.

Test tracks pose some added problems. To maintain neutral steer, they require parabolic cross sections. Normal screeds on paving machines cannot be configured to the required shape. This limitation necessitates the use of a screed with several sections. These special screeds usually have a break at 2-foot (0.6-meter) intervals. Compaction of the pavement on these parabolic sections also poses a problem. Either a roller with split drums or a series of smaller, narrow rollers are required.

Consideration should be given to diamond grinding the underlying layer before placing the surface course, eliminating the need to solve roughness problems with just the final surface.

Joints are very important and must be properly constructed to ensure good smoothness and proper density. The number of transverse construction joints should be minimized, and where production allows, a single paving lane should extend one full lap around the track. All joints should be tack-coated.

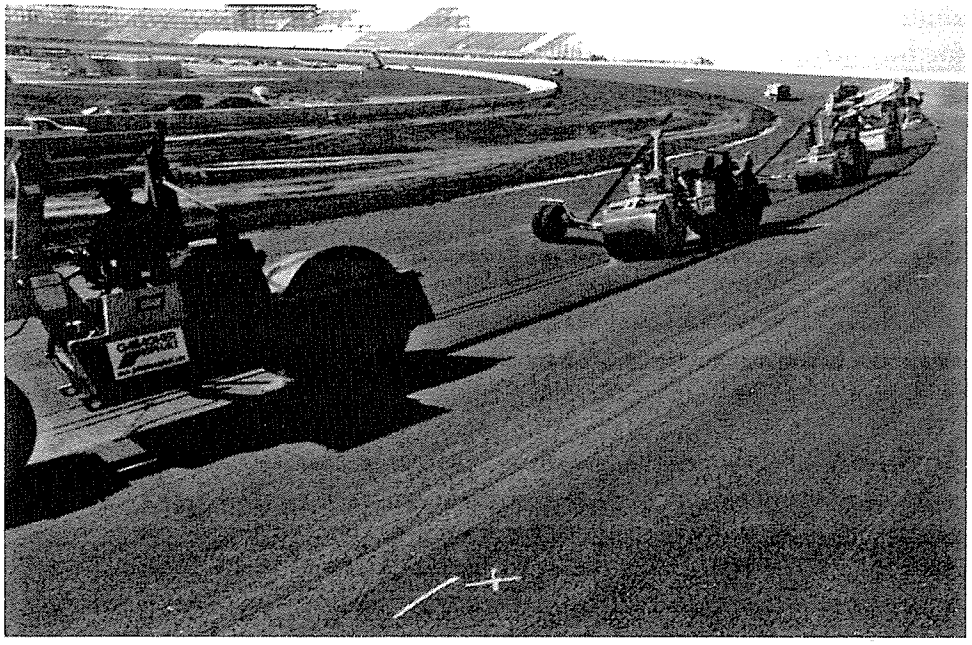


Figure 15.17 Roller Compacting HMA on Race Track

Good bond between all layers is very important. Tack coats on the very steep slopes present a problem and may have to be applied using a hand wand.

15.11 Porous Pavements

15.11.1 Introduction

A porous asphalt pavement can be used to reduce the runoff from parking lots and paved roads. A porous asphalt pavement consists of a permeable asphalt surface placed over a granular working platform on top of a reservoir of large single size stone. The asphalt surface is an open-graded friction course designed using the principles in Chapter 4, "Hot Mix Asphalt (HMA) Design." The lower reservoir of large single-size stone has sufficient voids to create adequate storage capacity. A properly designed porous asphalt pavement can provide a solution to storm-water runoff problems as well as groundwater table recharge. The National Asphalt Pavement Association's *Design, Construction, and Maintenance Guide for Porous Asphalt Pavements, Information Series 131*, contains detailed information on the design and construction of porous asphalt pavements. The following summarizes that information.

15.11.2 General Design Principles

A porous asphalt pavement consists of a porous asphalt course (usually an OGFC), a top filter course (of ½-inch [12.5 millimeters] aggregate), a reservoir course of aggregate (1½–3 inches; 37.5–75 millimeters thick), a filter fabric and the existing soil or subgrade material. Figure 15.18 shows a typical porous pavement. The site conditions, such as soil characteristics, local topography, climate, and traffic are the key items in determining the applicability of a porous pavement for a specific site. In addition, local storm-water regulations must be followed. The porous pavement should be designed for a 25-year/24-hour storm event. Other environmental factors must be considered. Frost penetration must be evaluated in determining the reservoir course thickness. Porous pavements are not recommended for barren areas subject to moderate to high winds because of the presence of blowing dust in those areas. Dust, sand, and salt used for snow and ice removal should be kept off a porous pavement because they can clog the voids in the porous pavement surface.

Soil and Subgrade Conditions

The subgrade on which the porous system is placed is a major factor in the success of a porous pavement. A thorough soil investigation must be conducted, including determining infiltration measurements performed at the approximate bottom of the porous pavement structure. These measurements can include simple percolation tests. These tests should be

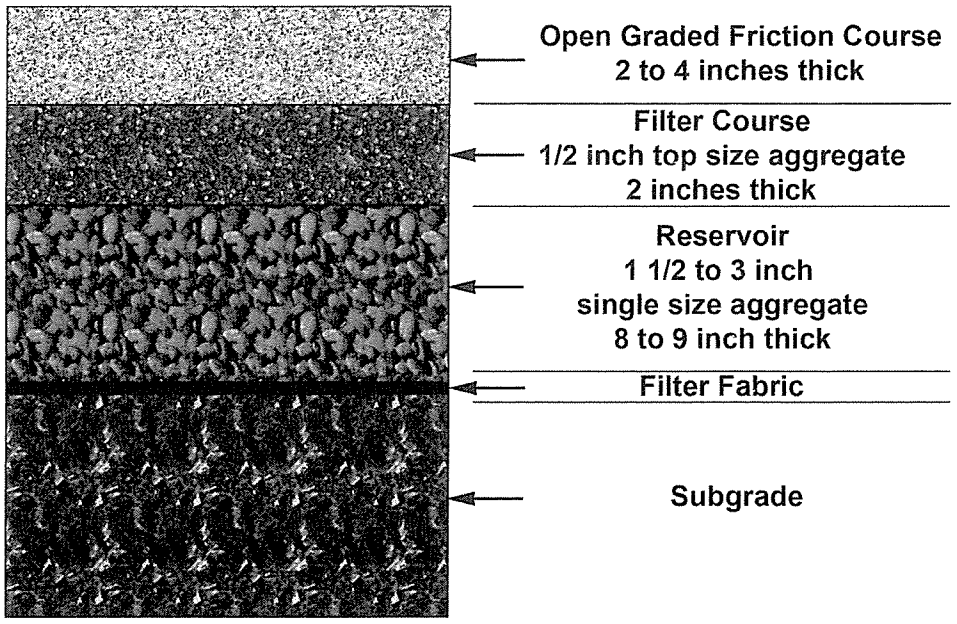


Figure 15.18 A Typical Porous Pavement System

performed at multiple locations throughout the site. It is suggested that the subgrade soils should have a minimum infiltration rate of 0.50 inch (12.5 millimeters) per hour for a full ex-filtration system. The goal is to design the system so that the water will drain within a 24- to 72-hour period from the retention reservoir.

Design of the Porous Pavement

The porous asphalt course consists of an OGFC approximately 2 to 4 inches (50 to 100 millimeters) thick. The OGFC should be designed following the procedures discussed in Chapter 4. The OGFC is placed on a 2-inch (50 millimeters) filter course of ½-inch (12.5 millimeters), single-sized crushed aggregate. A primary purpose of this course is to protect the reservoir course from disturbance when the paving machine places the OGFC. Without this firm paving platform, the single-size aggregate used for the reservoir layer might move under the paver.

The reservoir course consists of a single-size stone. The depth is a function of the storage volume, the structural needs of the pavement and the frost depth. A typical thickness for this layer is 8–9 inches (200–225 millimeters). The stone has a maximum top size of 1½–3 inches (37.5–75 millimeters). This gradation will provide the layer with approximately 40 percent air voids, which will provide for runoff storage. A 9-inch (225 millimeters) layer will provide a little over 3½ inches (88 millimeters) of runoff storage. If the reservoir is placed over marginal soils (infiltration rates of 0.25–0.50 inch/0.0625–0.125 millimeter per hour), the reservoir thickness must be increased. The porous pavement structure (when considering the reservoir depth and infiltration rate of the soil) should be

designed to drain completely within 24 hours. This feature allows the underlying soils to dry out between storms and thus preserves capacity for the next storm. Recommendations are that the thickness of the porous pavement structure exceed the depth of the frost penetration in the area where the porous pavement is being placed and that the pavement should be at least 2 feet (0.6 meter) above bedrock and 3 feet (1 meter) above the water table.

At the bottom of the porous pavement, a filter fabric should be placed to prevent fines from migrating into the layers above it. The fabric will also provide some stability for construction of the reservoir layer.

15.11.3 Construction

Constructing the porous pavement near the end of any other construction work prevents other construction debris from clogging the porous surface. The site area should be protected from excessive heavy equipment operating on the surface, which compacts the soil and reduces permeability. The following is a summary of the recommended construction sequence:

1. Excavate the subgrade using either track equipment or vehicles with over-sized tires to reduce the amount of compaction of the subgrade.
2. Place a filter fabric on the uncompacted subgrade.
3. Place 1½ to 3 inches (37.5 to 75 millimeters) of rock on the filter fabric in lifts to the design thickness, keeping construction equipment off the fabric. It should be lightly compacted with plate compactors or light rollers.
4. Place a thin 1- to 2-inch (25 to 50 millimeters) layer of ½-inch (12.5 millimeters) crushed aggregate over the rock layer.
5. Place a 2- to 4-inch (50 to 100 millimeters) layer of an OGFC. The OGFC should be designed as discussed in Chapter 4, "Hot Mix Asphalt (HMA) Design." A polymer-modified asphalt should be considered.

15.11.4 Maintenance

The porous pavement should be inspected periodically to check for surface ponding, which indicates possible clogging. The porous surface can be vacuumed, flushed or jet washed to help maintain the pavement's porosity.

15.12 Colored Asphalt

Although not widely done, asphalt pavements can be colored something other than the usual shades of black and grey. The first coloring method is to use a natural-color aggregate and the wearing (surface) course. As the asphalt binder wears way from the surface with traffic, the aggregate is exposed, revealing its natural color. A second way to color asphalt is to use an additive in the asphalt binder such as an iron oxide (rust) to give it a reddish tint. Green, yellow, and orange are also possible with different ores. Other colors require different metal compounds. A special synthetic binder (resin) that contains no asphaltenes has been used because it colors more readily. By tinting the asphalt binder, the hot mix asphalt pavement is colored full depth, so there are no surface wear-off concerns. A third way to color asphalt is to coat the surface with a material that penetrates the voids and bonds well to an asphalt pavement, such as an epoxy-fortified acrylic emulsion. Many colors are available with these types of surface coatings. Care should be taken that these surface coatings do not negatively affect surface friction if the pavement is used for vehicular traffic. One possible disadvantage of these systems is that they may wear off with time and need to be renewed. The second and third options require the use of specialty products, and more information about them can be obtained by contacting individual manufacturers.

15.13 Warm Mix Asphalt Pavements

New technologies have been developed that allow asphalt mixtures to be produced and placed at significantly lower temperatures. These technologies are called warm mix asphalt (WMA). With these technologies, temperatures can be reduced by as much as 30 percent but still allow the asphalt binder to adequately coat the aggregate during mixing at the plant and achieve the desired workability at the paving site. These methods reduce the viscosity and increase the workability of a given asphalt binder at a given temperature.

15.13.1 Potential Benefits

The following benefits in reducing mix production and paving temperatures through WMA are most often mentioned:

- Decreasing the energy and fuel consumed to make HMA
- Reducing possible emissions and odors from plants
- Improving the working conditions at the plant and paving site
- Allows for cold-temperature paving

Reducing emissions is especially critical in and around nonattainment areas such as large metropolitan areas and cities that have tight air quality restrictions.

Other potential benefits of WMA are less recognized. Some contractors and agencies are exploring the use of WMA in order to extend the paving season in cool weather. Depending on location and climate, paving projects are typically prohibited after a certain calendar date because colder temperatures lessen the time available for the rolling operation to compact the mix to the desired in-place density. Through WMA technology, mixes can remain workable at cooler temperatures, increasing the window of opportunity for compaction. This may make WMA a feasible option for those end-of-season projects whose completions before winter without compromising quality are so important.

Early indications are that WMA technology can provide all these benefits without sacrificing the long-term performance of the asphalt pavement. However, before its widespread adoption, WMA technology must prove that is provides the same long-term performance as regular HMA.

15.13.2 Various WMA Technologies

WMA was first developed and used in Europe and came to the United States in 2002. The following sections provide a brief overview of the various WMA processes. Many are proprietary and not available in all geographic

areas. Specifying engineers and agencies should check with the local binder supplier to confirm what processes are available in their area.

Organic (Wax-Like) Additives

Special organic wax-like additives are mixed with the binder to reduce viscosity at temperatures above about 212°F (100°C) and then increase the viscosity at temperatures below about 212°F (100°C). The shift from viscosity reduction to viscosity increase in the binder occurs near the congealing point of the wax.

Mineral Additives

A number of minerals that have water bound internally have been used as additives in order to introduce water into an HMA mixture. The water is liberated gradually over time, resulting in an almost continuous foaming action. This foaming action of the liquid binder acts as an extender and lubricant that enables the mix to be workable at lower temperatures. Production temperatures can be reduced to the 266°–293°F (130°–145°C) range. The materials are added in a manner that produces up to 0.6 percent water by weight of total mix.

Low-Energy Asphalt

Another WMA process, developed in France, uses wet fine aggregate. This process is called Low-Energy Asphalt (LEA). A key to the energy savings in this process is that it takes five times more energy to turn water into steam than it takes to heat the aggregate from 32°F to 212°F (0°–100°C). Fine aggregates tend to have the highest moisture contents. In the LEA process, the coarse-aggregate fraction is dried and heated to a temperature of approximately 300°F (149°C). A specially formulated surfactant is then added to the binder. The coarse aggregates are then coated with the binder, which results in a thick film coating because the fine aggregate has not yet been added. The wet fine aggregate is then mixed with the asphalt-coated coarse aggregate, resulting in a foaming action that aids in the coating of the fine aggregate.

Foam Processes

Foamed asphalt is created by injecting cold water, typically 1–5 percent, into the hot asphalt binder using special equipment. The water rapidly evaporates, producing a large volume of foam, which slowly collapses. Foamed asphalt differs from foaming mineral additives in that the water and asphalt binder are directly mixed together in the presence of compressed air. The expanded foamed binder is then injected directly into the mixing chamber containing the aggregate. In a variation of the foamed process, the aggregates are heated and coated with a very soft binder. A very hard asphalt binder is then foamed into the mixing chamber. The combined soft and hard binder produces the intended binder grade.

Emulsion-Based Process

An emulsion-based process uses a chemistry package that includes materials to improve workability, adhesion promoters, and emulsification